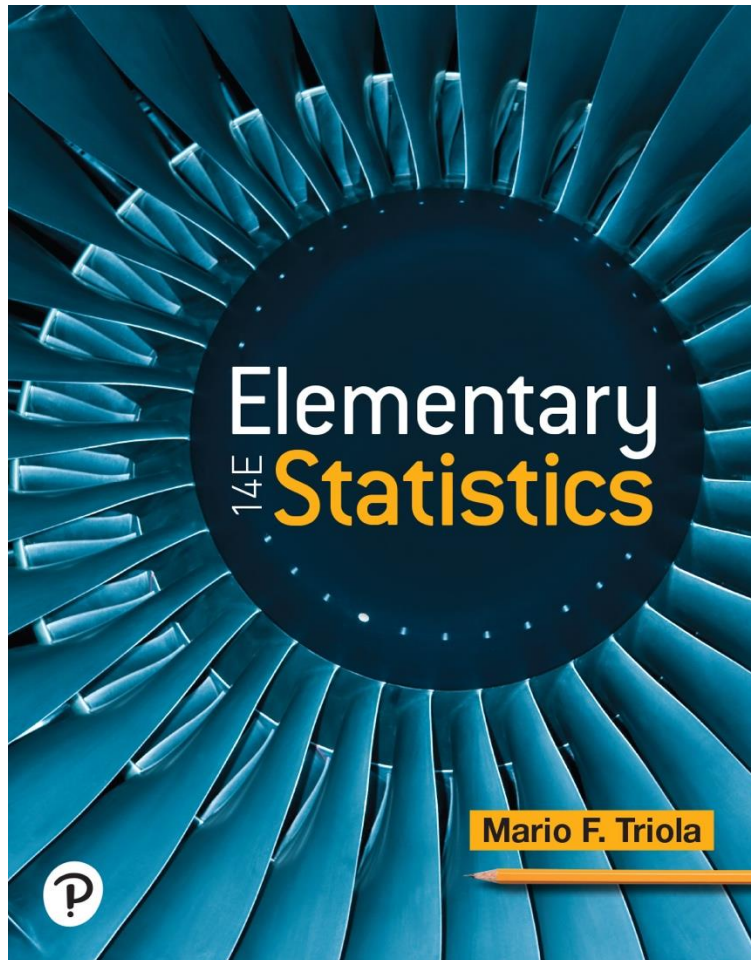


Elementary Statistics

Fourteenth Edition



Chapter 5

Probability Distributions

Probability Distributions

5-1 Probability Distributions

5-2 Binomial Probability Distributions

5-3 Poisson Probability Distributions

Key Concept

This section introduces the concept of a **random variable** and the concept of a **probability distribution**.

We show how to find the important parameters of mean, standard deviation, and variance for a probability distribution.

Most importantly, we describe how to determine whether outcomes are **significantly low** or **significantly high** or neither.

Basic Concepts of Probability Distribution (1 of 4)

- Random Variable
 - A **random variable** is a variable (typically represented by x) that has a single numerical value, determined by chance, for each outcome of a procedure.

Basic Concepts of Probability Distribution (2 of 4)

- Probability Distribution
 - A **probability distribution** is a description that gives the probability for each value of the random variable. It is often expressed in the format of a table, formula, or graph.

Basic Concepts of Probability Distribution (3 of 4)

- Discrete Random Variable
 - A **discrete random variable** has a collection of values that is finite or countable. (If there are infinitely many values, the number of values is countable if it is possible to count them individually, such as the number of tosses of a coin before getting heads.)

Basic Concepts of Probability Distribution (4 of 4)

- Continuous Random Variable
 - A **continuous random variable** has infinitely many values, and the collection of values is not countable. (That is, it is impossible to count the individual items because at least some of them are on a continuous scale, such as body temperatures.)

Probability Distribution Requirements (1 of 2)

Every probability distribution must satisfy each of the following three requirements.

1. There is a **numerical** (not categorical) random variable x , and its numerical values are associated with corresponding probabilities.
2. $\sum P(x) = 1$ where x assumes all possible values. (The sum of all probabilities must be 1, but sums such as 0.999 or 1.001 are acceptable because they result from rounding errors.)

Probability Distribution Requirements (2 of 2)

3. $0 \leq P(x) \leq 1$ for every individual value of the random variable x . (That is, each probability value must be between 0 and 1 inclusive.)

Example: Coin Flipping (1 of 3)

Assume that a coin has a 50 / 50 chance of landing heads or tails. Let's consider two flips, with the following random variable:

x = number of heads when flipping a coin twice

The above x is a random variable because its numerical values depend on chance.

x: Number of Heads in Two Flips	$P(x)$
0	0.25
1	0.50
2	0.25



Example: Coin Flipping (2 of 3)

With two flips, the number of heads can be 0, 1, or 2, and the table is a probability distribution because it gives the probability for each value of the random variable x and it satisfies the three requirements listed earlier:

1. The variable x is a **numerical** random variable, and its values are associated with probabilities.
2. $\sum P(x) = 0.25 + 0.50 + 0.25 = 1$
3. Each value of $P(x)$ is between 0 and 1.

Example: Coin Flipping (3 of 3)

The random variable x in the table is a discrete random variable, because it has three possible values (0, 1, 2), so this satisfies the requirement of being finite.

x: Number of Heads in Two Flips	$P(x)$
0	0.25
1	0.50
2	0.25

Probability Formula

A probability distribution could also be in the form of a formula. Consider the formula

$$P(x) = \frac{1}{2(2-x)!x!} \quad (\text{where } x \text{ can be } 0, 1 \text{ or } 2).$$

We find that $P(0) = 0.25$, $P(1) = 0.50$, and $P(2) = 0.25$. The probabilities found using this formula are the same as those in the table.

Example: Software Piracy (1 of 2)

The table lists countries along with the proportion of unlicensed software in each country (based on data from Business Software Alliance). Does the table describe a probability distribution?

Country	Proportion of Unlicensed Software
United States	0.17
China	0.70
India	0.58
Russia	0.64
Total	2.09

Example: Software Piracy (2 of 2)

Solution

The table violates the first requirement because x is not a **numerical** random variable. The “values” of x are categorical data, not numbers.

The table also violates the second requirement because the sum of the probabilities is 2.09, but that sum should be 1.

Because the three requirements are not all satisfied, we conclude that the table does **not** describe a probability distribution.

Parameters of a Probability Distribution (1 of 3)

Remember that with a probability distribution, we have a description of a **population** instead of a sample, so the values of the mean, standard deviation, and variance are **parameters**, not statistics.

The mean, variance, and standard deviation of a discrete probability distribution can be found with the following formulas:

Parameters of a Probability Distribution (2 of 3)

- **Mean, μ ,** for a probability distribution

$$\mu = \sum [x \cdot P(x)]$$

- **Variance, σ^2 ,** for a probability distribution

$$\sigma^2 = \sum [(x - \mu)^2 \cdot P(x)] \quad \text{(This format is easier to understand.)}$$

- **Variance, σ^2 ,** for a probability distribution

$$\sigma^2 = \sum [x^2 \cdot P(x)] - \mu^2 \quad \text{(This format is easier for manual calculations.)}$$

Parameters of a Probability Distribution (3 of 3)

- **Standard deviation, σ** , for a probability distribution

$$\sigma = \sqrt{\sum [x^2 \cdot P(x)] - \mu^2}$$

Expected Value

- Expected Value
 - The **expected value** of a discrete random variable x is denoted by E , and it is the mean value of the outcomes, so $E = \mu$ and E can also be found by evaluating $\sum[x \cdot P(x)]$.

Example: Finding the Mean, Variance, and Standard Deviation (1 of 5)

The table describes the probability distribution for the number of heads in two flips. Find the mean, variance, and standard deviation for the probability distribution described.

x: Number of Heads in Two Flips	$P(x)$
0	0.25
1	0.50
2	0.25

Example: Finding the Mean, Variance, and Standard Deviation (2 of 5)

x	$P(x)$	$x \cdot P(x)$	$(x - \mu)^2 \cdot P(x)$
0	0.25	$0 \cdot 0.25 = 0.00$	$(0 - 1.0)^2 \cdot 0.25 = 0.25$
1	0.50	$1 \cdot 0.50 = 0.50$	$(1 - 1.0)^2 \cdot 0.50 = 0.00$
2	0.25	$2 \cdot 0.25 = 0.50$	$(2 - 1.0)^2 \cdot 0.25 = 0.25$
Total		1.00 $\mu = \sum[x \cdot P(x)]$	0.50 $\sigma^2 = \sum[(x - \mu)^2 \cdot P(x)]$

Solution

The two columns at the left describe the probability distribution. The two columns at the right are for the purposes of the calculations required.

Example: Finding the Mean, Variance, and Standard Deviation (3 of 5)

x	$P(x)$	$x \cdot P(x)$	$(x - \mu)^2 \cdot P(x)$
0	0.25	$0 \cdot 0.25 = 0.00$	$(0 - 1.0)^2 \cdot 0.25 = 0.25$
1	0.50	$1 \cdot 0.50 = 0.50$	$(1 - 1.0)^2 \cdot 0.50 = 0.00$
2	0.25	$2 \cdot 0.25 = 0.50$	$(2 - 1.0)^2 \cdot 0.25 = 0.25$
Total		1.00 $\mu = \sum[x \cdot P(x)]$	0.50 $\sigma^2 = \sum[(x - \mu)^2 \cdot P(x)]$

Solution

$$\text{Mean: } \sum[x \cdot P(x)] = 1$$

$$\text{Variance: } \sum[(x - \mu)^2 \cdot P(x)] = 0.5$$

Example: Finding the Mean, Variance, and Standard Deviation (4 of 5)

x	$P(x)$	$x \cdot P(x)$	$(x - \mu)^2 \cdot P(x)$
0	0.25	$0 \cdot 0.25 = 0.00$	$(0 - 1.0)^2 \cdot 0.25 = 0.25$
1	0.50	$1 \cdot 0.50 = 0.50$	$(1 - 1.0)^2 \cdot 0.50 = 0.00$
2	0.25	$2 \cdot 0.25 = 0.50$	$(2 - 1.0)^2 \cdot 0.25 = 0.25$
Total		1.00 $\mu = \sum[x \cdot P(x)]$	0.50 $\sigma^2 = \sum[(x - \mu)^2 \cdot P(x)]$

Solution

The standard deviation is the square root of the variance, so

$$\text{Standard deviation: } \sigma = \sqrt{0.5} = 0.707107 = 0.7 \text{ (rounded)}$$

Example: Finding the Mean, Variance, and Standard Deviation (5 of 5)

Interpretation

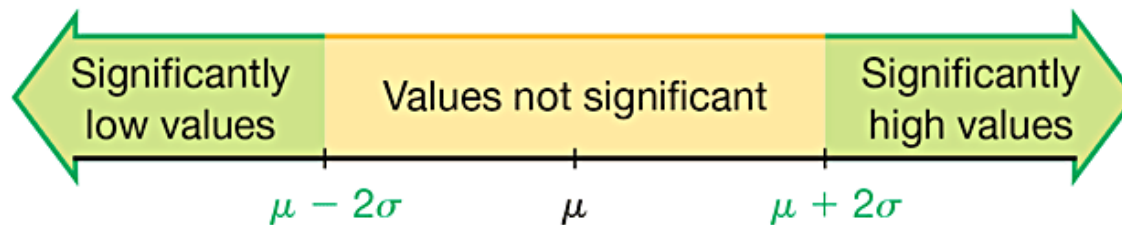
In two flips, the mean number of heads is 1.0 head, the variance is 0.5 head², and the standard deviation is 0.7 head.

Also, the expected value for the number of heads in two flips is 1.0 head, which is the same value as the mean. If we were to collect data on a large number of trials with two flips in each trial, we expect to get a mean of 1.0 head.

Identifying Significant Results with the Range Rule of Thumb

Range Rule of Thumb for Identifying Significant Values

- **Significantly low** values are $(\mu - 2\sigma)$ or lower.
- **Significantly high** values are $(\mu + 2\sigma)$ or higher.
- **Values not significant:** Between $(\mu - 2\sigma)$ and $(\mu + 2\sigma)$.



Example: Identifying Significant Results with the Range Rule of Thumb (1 of 3)

We found that for two flips, the mean number of heads is $\mu = 1.0$ head and the standard deviation is $\sigma = 0.7$ head. Use those results and the range rule of thumb to determine whether 2 heads in 2 flips is significantly high.

Example: Identifying Significant Results with the Range Rule of Thumb (2 of 3)

Solution

Using the range rule of thumb, the outcome of 2 heads is significantly high if it is greater than or equal to $(\mu + 2\sigma)$. With $\mu = 1.0$ head $\sigma = 0.7$ head, we get

$$(\mu + 2\sigma) = 1 + 2(0.7) = 2.4 \text{ heads}$$

Significantly high numbers of heads are 2.4 and above.

Example: Identifying Significant Results with the Range Rule of Thumb (3 of 3)

Interpretation

Based on these results, we conclude that 2 heads is not a significantly high number of heads (because 2 is not greater than or equal to 2.4).

Identifying Significant Results with Probabilities: (1 of 2)

- Significantly high number of successes:
 - x successes among n trials is a **significantly high** number of successes if the probability of x or more successes is 0.05 or less. That is, x is a significantly high number of successes if $P(x \text{ or more}) \leq 0.05$.

The value 0.05 is not absolutely rigid. Other values, such as 0.01, could be used to distinguish between results that are significant and those that are not significant.

Identifying Significant Results with Probabilities: (2 of 2)

- Significantly low number of successes:
 - x successes among n trials is a **significantly low** number of successes if the probability of x or fewer successes is 0.05 or less. That is, x is a significantly low number of successes if $P(x \text{ or fewer}) \leq 0.05$.

The value 0.05 is not absolutely rigid. Other values, such as 0.01, could be used to distinguish between results that are significant and those that are not significant.

The Rare Event Rule for Inferential Statistics

If, under a given assumption, the probability of a particular outcome is very small and the outcome occurs **significantly less than or significantly greater than** what we expect with that assumption, we conclude that the assumption is probably not correct.